Team Name: DeepEpoch KDSH-2026

Overall Approach

Objective and Formulation

The primary objective of this work is to determine whether a hypothetical backstory is globally consistent with a long-form narrative or if it contradicts the established events, facts, or logical structure of the story. To address this, the task is examined as a Natural Language Inference (NLI) problem. In this setup, a segment of the narrative serves as the *premise* while the backstory functions as the *hypothesis*. This formulation allows the system to move beyond shallow keyword overlap and explicitly reason about logical relationships, distinguishing between entailment, neutrality, and contradiction.

Methodological Approach

To handle the complexity of long-form text and the input limitations of transformer models, the system employs the following pipeline:

- **Text Segmentation:** Since full novels exceed model input limits, the text is divided into smaller, fixed-size chunks (400 tokens) that fit within constraints while preserving local narrative coherence.
- **Context Retrieval:** To avoid the inefficiency of evaluating every segment, the system utilises TF-IDF vectorization combined with cosine similarity. This identifies the **top-k most semantically relevant narrative chunks** to ensure broader context coverage.
- **Deep Semantic Processing:** The retrieved narrative chunk and the backstory are jointly processed by a pretrained RoBERTa-large model that has been fine-tuned on the Multi-Genre Natural Language Inference (MNLI) dataset.

Classification and Outcome

The model outputs a probabilistic assessment regarding the relationship between the texts. These predictions are mapped to binary task labels: **contradictions** are classified as inconsistent, while **entailment or neutral** cases are classified as consistent. The result is a fully automated, computationally efficient, and reproducible pipeline that effectively combines information retrieval techniques with deep semantic reasoning to evaluate narrative consistency.

Installation and Setup

1)Run ‘pip install -r requirements.txt’ in your bash terminal to install dependencies.

2)Make sure you have provided the path of the backstory and the novel correctly.

3) Run ‘run.py’ in your bash terminal to activate the main engine.

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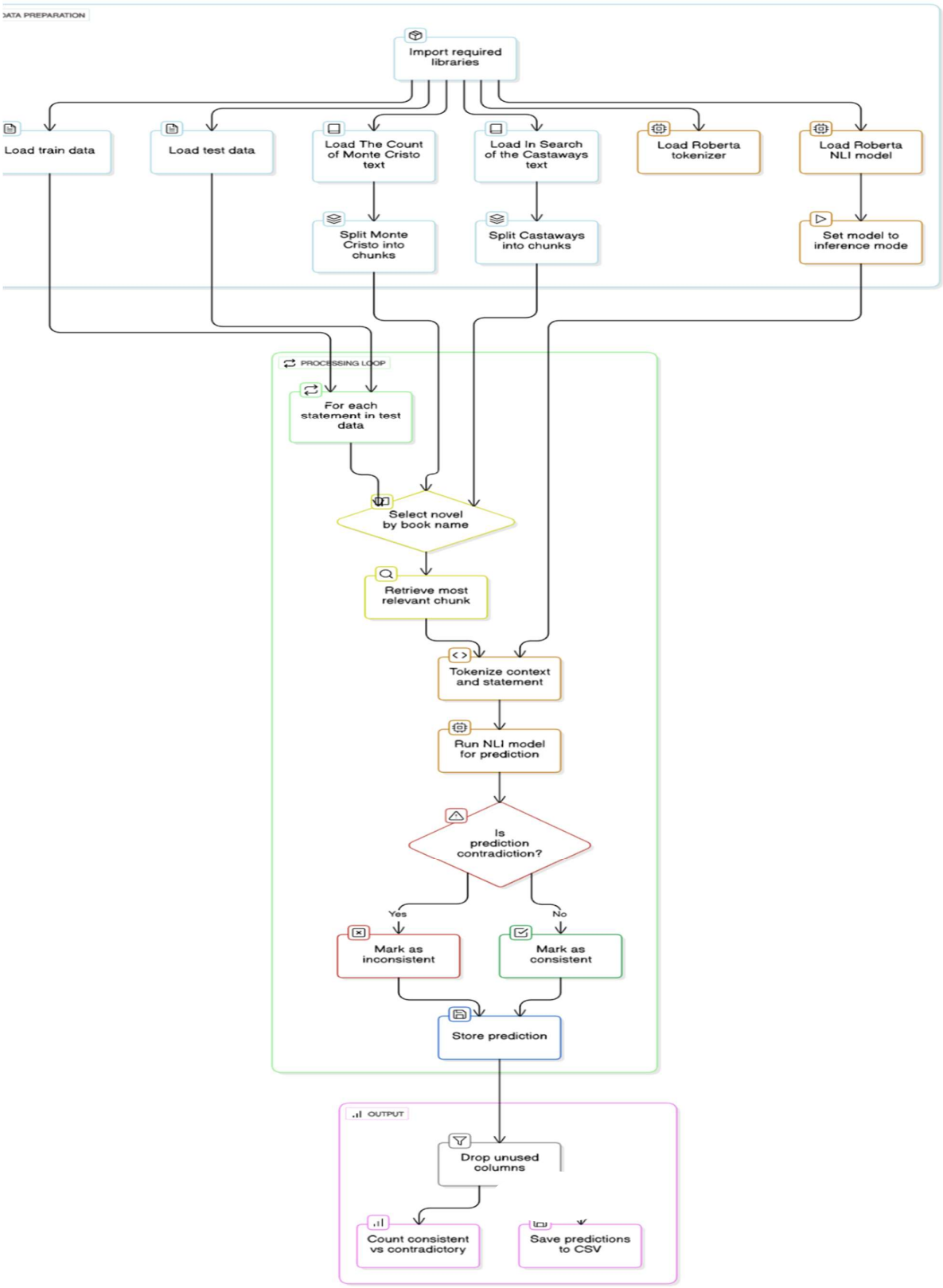
FLOWCHART



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Handling Long Context

A central challenge in this task is the effective processing of long-form narrative texts, as the provided novels consist of tens of thousands of words, far exceeding the input length limitations of transformer-based language models. Models such as 'RoBERTa' are typically restricted to a maximum of 512 tokens per input sequence, making it infeasible to directly analyse an entire novel in a single inference step. Without addressing this limitation, important narrative information would either be truncated or ignored, leading to unreliable consistency judgments. The internal working of transformers is shown in Figure 1.

To overcome this challenge, a chunking-based strategy is adopted. Each novel is segmented into smaller, fixed-size chunks of approximately 400 words. This chunk size is chosen to balance two competing objectives: retaining sufficient narrative coherence within each segment while ensuring compatibility with the transformer's maximum input constraints after tokenisation. By dividing the novel into manageable units, the system is able to represent the complete narrative in a form suitable for downstream processing.

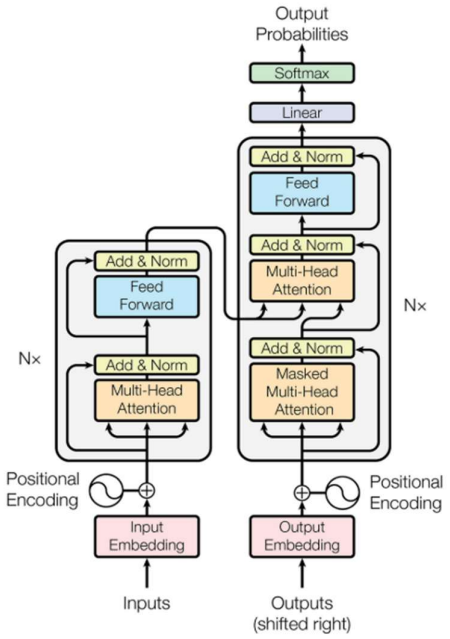


Figure 1: The Transformer - model architecture.

However, evaluating every chunk for each backstory would be computationally expensive and unnecessary, as most narrative segments are irrelevant to a given statement. To address this, an information retrieval step is introduced to identify the most relevant chunk selectively. Using TF-IDF vectorisation, both the backstory and the narrative chunks are converted into **numerical representations** that capture the relative importance of words within the corpus. Cosine similarity is then computed between the backstory vector and each chunk vector to measure semantic relevance. The **top-k (k=3) chunks** with the highest similarity scores are selected as contextual evidence. This multi-chunk retrieval reduces the risk of missing critical evidence that might appear in the second or third most relevant segment.

This retrieval-based approach ensures that the reasoning model focuses only on the most informative portion of the narrative, significantly reducing noise and computational cost. While this method does not guarantee perfect coverage of globally distributed evidence, it provides an effective and scalable solution for handling long contexts in resource-constrained inference settings.

Context Retrieval Method

Given the extensive length of narrative texts, evaluating every segment for a specific backstory is inefficient and prone to error. Most parts of a novel are unrelated to any single specific statement, and including this irrelevant context introduces noise that can degrade reasoning performance. To address this, an explicit retrieval mechanism is used to filter the text and isolate the most relevant segment before logical evaluation begins.

Retrieval Methodology

The system employs a classical but robust information retrieval pipeline:

- **TF-IDF Vectorisation:** The process utilises Term Frequency–Inverse Document Frequency (TF-IDF) to convert text into numerical vectors. This method balances **Term Frequency** (how often a word appears in a chunk) with **Inverse Document Frequency** (which down-weights common words to emphasise unique, informative terms). This results in sparse vectors that capture semantic content while minimizing non-informative noise.
- **Cosine Similarity Scoring:** To match a backstory with the narrative, the system calculates the cosine similarity between the backstory's vector and the vectors of all narrative chunks. This metric measures the angle between vectors in high-dimensional space, providing a normalised score that reflects topical alignment independent of the text length.
- **Context Selection:** The narrative chunk with the highest similarity score is selected as the specific contextual evidence.

Impact and Efficiency

By ensuring the logical inference model only processes the most lexically similar portion of the text, this strategy significantly reduces computational overhead. Although it relies on lexical overlap rather than deep implicit reasoning, it provides a highly efficient filtering mechanism. This improves the overall scalability of the system and ensures that the model is not overwhelmed by irrelevant information during long-context consistency tasks.

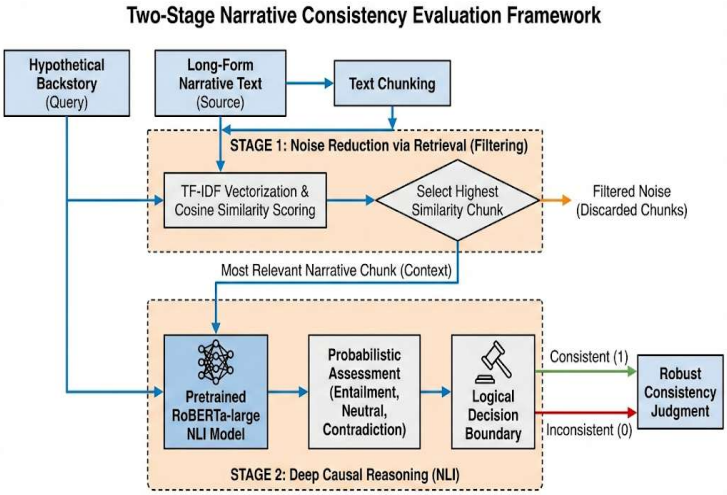
Distinguishing Causal Signals from Noise

In long-form narratives, not all text contributes equally to logical consistency. Much of a story consists of descriptive details, stylistic flourishes, or unrelated events that function as "background noise" rather than driving the causal structure. Distinguishing these meaningful causal signals from irrelevant information is essential for accurate evaluation. To achieve this, the proposed approach employs a two-stage filtering and reasoning process.

Two-Stage Filtering and Reasoning

This system separates signal from noise through the following mechanisms:

- **Noise Reduction via Retrieval:** The first stage filters out unrelated narrative content. By utilizing TF-IDF vectorization and cosine similarity, the system isolates the **top-most relevant narrative chunks** for each backstory. In the reasoning stage, **an 'Any-Contradiction' aggregation logic** is applied: if the NLI model detects a contradiction in *any* of the retrieved chunks, the backstory is flagged as inconsistent. This significantly reduces false positives caused by missing context. This effectively discards large portions of text—such as scene descriptions or distinct subplots—that are unlikely to contain causally relevant information.
- **Deep Causal Reasoning:** The second stage employs a pretrained RoBERTa-large NLI model to move beyond surface-level similarity. Unlike simple keyword matching, this model detects deep logical relationships, identifying explicit and implicit inconsistencies (e.g., actions contradicting character motivations or outcomes violating prior conditions).
- **Logical Decision Boundary:** The system adopts a specific threshold for consistency: only **contradiction** predictions are flagged as inconsistent. **Neutral** predictions are treated as consistent. This design focuses on detecting active logical conflicts rather than penalizing the model for missing or unstated information, thereby minimizing false negatives.



Robust Consistency Judgments

By combining distinct retrieval and reasoning stages, the system effectively discriminates between causally significant signals and narrative noise. This ensures that the evaluation focuses on genuine causal violations, resulting in judgments that are both robust and interpretable

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Results Format and Reproducibility

To ensure ease of evaluation and consistency across multiple runs, the system generates final outputs in a structured and reproducible format. The workflow is fully automated, producing a deterministic output that adheres strictly to submission guidelines, which allows for seamless automated evaluation without the need for manual post-processing.

Format and Labelling Details

The specific characteristics of the output and system design include:

- **Strict CSV Schema:** The final predictions are stored in a comma-separated values (CSV) file named result.csv. This file contains exactly two columns: the unique **story identifier** and the corresponding **binary label**. No additional metadata or text is included, ensuring compatibility with automated scoring scripts.
- **Binary Labelling Scheme:** The system uses a clear binary mapping for the label column. A value of **1** denotes consistency (the backstory does not contradict the narrative), while a value of **0** indicates an inconsistency (a logical contradiction exists).
- **Deterministic Reproducibility:** The pipeline is designed to be fully reproducible. It uses publicly available pretrained models and operates in inference mode without stochastic training steps, ensuring that the same input data always yields the same CSV output as it has been asked to attach.
- **Modular Architecture:** The system is built with independent components for chunking, retrieval, inference, and result generation. This allows individual stages to be verified or replaced by other researchers without disrupting the overall system logic.

Sample view of rows in "*Result.csv*"

'Story_id' in code is reffered as 'id'(in csv file)

```
id,prediction
95,0
136,0
59,0
60,1
124,0
```

System Verification

This modular and transparent design ensures that the pipeline is not only computationally efficient but also fully extensible. By relying on a minimal, standardised output format and public models, the approach allows other evaluators to easily reproduce results or integrate alternative reasoning models while maintaining strict compatibility with the evaluation framework

Limitations and Failure Cases

While the proposed approach effectively combines information retrieval and natural language inference for consistency evaluation, it operates under specific constraints that may lead to failure cases in complex scenarios. Identifying these limitations is essential for correctly interpreting results and establishing a roadmap for future improvements.

Key Challenges

The primary limitations of the current pipeline include:

- **Localised Context Constraints:** While the Top-K (K=3) retrieval strategy broadens context coverage, the system still processes each chunk independently. It cannot synthesise information across widely separated chapters (e.g., relating a fact in Chapter 1 to an event in Chapter 20) unless they share significant lexical features. Consequently, contradictions relying on distributed evidence without keyword overlap may still be misclassified as consistent
- **Reliance on Lexical Similarity:** The use of TF-IDF retrieval depends heavily on shared vocabulary. This method may fail when contradictions are implicit or expressed using paraphrased language. If the relevant narrative text lacks keyword overlap with the backstory, the retrieval mechanism may select an unrelated chunk, leading to incorrect inference.
- **Domain Specificity of the Model:** The RoBERTa-large model, while powerful for general sentence-level logic, is not specifically fine-tuned for literary narratives. It may struggle with nuanced contradictions involving long-term character development, temporal shifts, or implied causal relationships that require a broader understanding of the story arc.
- **Risks of Neutral Classifications:** The decision to treat "neutral" predictions as "consistent" prioritizes the detection of active contradictions. However, this may introduce false positives; a neutral label often indicates a lack of evidence rather than true consistency. Consequently, the system may overestimate consistency when the retrieved context is incomplete.
- **Computational Trade-offs:** To ensure scalability and efficiency, the system avoids computationally expensive techniques like multi-context aggregation or dense vector retrieval. This design choice trades off deeper, global reasoning capabilities for practical performance.

Implications for Future Work

These limitations highlight the trade-off between computational efficiency and deep semantic understanding. While the current pipeline is robust for direct contradictions, future iterations could improve accuracy through multi-chunk reasoning, semantic embedding-based retrieval (to capture implicit meaning), or domain-specific fine-tuning on literary texts.

Conclusion

This work presents a systematic and reproducible approach for evaluating the global consistency of hypothetical backstories with long-form narrative texts. By framing the task as a Natural Language Inference problem, the proposed pipeline moves beyond surface-level text matching and enables logical reasoning about contradictions and narrative coherence. The combination of text chunking and context retrieval effectively addresses the challenge posed by the extreme length of the narratives, allowing the system to focus on the most relevant portions of the story while remaining computationally efficient.

The integration of a pretrained RoBERTa-large MNLI model provides a robust mechanism for detecting logical conflicts between backstories and narrative context. By mapping contradiction predictions to inconsistency and treating entailment or neutral cases as consistent, the system aligns closely with the task objective of identifying genuine narrative contradictions. The modular design of the pipeline further enhances reproducibility, interpretability, and extensibility, making it suitable for both evaluation and future research.

While the approach has certain limitations, particularly in handling distributed evidence and deeply implicit contradictions, it establishes a strong baseline for global narrative consistency evaluation. Overall, this work demonstrates that combining classical information retrieval techniques with transformer-based reasoning models offers an effective and scalable solution for consistency analysis in long-form textual narratives.